

Angle dependence of Brillouin scattering intensity in liquids

YE BO,^{1,2,†}  SHUAI YAO,^{1,2,†} YUN QI,^{1,2} AND FAN YANG^{1,2,*} 

¹Shanghai Institute of Optics and Fine Mechanics, Chinese Academy of Sciences, Shanghai, China

²University of Chinese Academy of Sciences, Beijing, China

[†]These authors contributed equally to this work.

*yang@siom.ac.cn

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We developed an off-axis Brillouin spectroscopy system with wide-angle tuning and polarization selection capabilities to study the angle-dependent intensity of Brillouin scattering. To compensate for variations in collection efficiency across different scattering angles, we standardized Brillouin signal intensities using Rayleigh scattering as a reference. Using a virtual imaged phased array (VIPA) spectrometer, we achieved fast, simultaneous, high-precision measurements of Rayleigh and Brillouin signals, reducing acquisition time to sub-second levels. We measured the Landau–Placzek ratio of water to be 0.015, an improvement in accuracy compared to previous reports and closer to the theoretical value of 0.01. Our results confirm that Brillouin and Rayleigh scattering share the same angle dependence, demonstrating that the dipole scattering model is well-suited for Brillouin scattering in liquids. This work establishes a simple yet robust framework for accurate spectral analysis and strengthens the applicability of Brillouin microscopy in biomechanical research.

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Spontaneous Brillouin scattering is an inelastic process resulting from the interaction between light and thermally induced phonons. The frequency shift provides information about acoustic velocity, stiffness, or longitudinal modulus, while the linewidth reflects the attenuation coefficient or viscosity. Recently, Brillouin microscopy has emerged as a powerful technique for probing cell and tissue biomechanics. Most systems are based on either backward scattering [1–3] or right-angle scattering geometries [4,5]. The properties of Brillouin scattering are closely related to the scattering angle θ , with the frequency shift $\omega \propto \sin(\theta/2)$ and the linewidth $\Gamma \propto \sin^2(\theta/2)$. So far, only a few studies have explored the angle dependence of Brillouin scattering. Fiore *et al.* probed the refractive index using Brillouin frequency shifts obtained in two different scattering angles [6], while Zhang *et al.* compared the scattering power collected in orthogonal and epi-configurations [7]. However, the intensity dependence on the angle has not been fully demonstrated, which is essential for Brillouin microscopy. Especially when high numerical aperture (NA) objectives are used, the broader

distribution of scattering vectors necessitates an accurate calculation of the entire spectrum. Although previous studies have investigated spectral broadening of spontaneous Brillouin scattering induced by NA of the objective [8], they have simply superimposed spectra across scattering angles while neglecting the angle dependence of the Brillouin scattering intensity. This oversight can lead to an inaccurate estimation of the spectral broadening effect from high NA objectives.

Direct measurement of the Brillouin scattering intensity accurately from different angles is challenging because the overlap volume of the excitation and collection beams, which dominates the collection efficiency, changes significantly with the scattering angle [9]. Nevertheless, since Rayleigh and Brillouin scattering signals are collected using the same optical system, we can leverage Rayleigh scattering signals to standardize the Brillouin scattering signals across different angles. This approach is applicable because the Rayleigh scattering theory, which has been well established since the 19th century [10], provides a solid foundation for understanding the scattering behavior and offers valuable insights for Brillouin scattering intensity measurements.

The relative intensities of Rayleigh to Brillouin scattering are known as the Landau–Placzek ratio. Previous studies measuring this ratio have been limited to right-angle scattering geometry, using Fabry–Perot interferometers to acquire the full spectrum through cavity scanning. This scanning approach not only requires long acquisition time but also makes it difficult to separate fluctuations in Rayleigh and Brillouin scattering from other noise sources, such as dust scattering or variations in laser power.

In this work, we designed an off-axis Brillouin spectroscopy system, which offers a wide range of angle tuning and allows for arbitrary polarization selection for both the input and scattered light, making it ideal for studying angle and polarization dependencies. Additionally, this approach eliminates most of the stray light, so no filters are needed in the system. For the first time, we used a virtual imaged phased array (VIPA) based Brillouin spectroscopy setup to measure the Landau–Placzek ratio in various scattering angles. Faster acquisition speeds and higher precision were achieved in our VIPA based system compared to previous works using the Fabry–Perot interferometer, as Rayleigh and Brillouin signals were measured simultaneously.

Theoretical analysis. The Landau–Placzek ratio is predicted to be as follows [11]:

$$\frac{I_c}{2I_B} = \frac{\beta_T - \beta_S}{\beta_S}, \quad (1)$$

where I_c and I_B are the Rayleigh and Brillouin components, respectively. β_T and β_S are the isothermal and adiabatic compressibilities, respectively.

This equation is generally applicable to light scattering in liquids, as confirmed by numerous studies [12,13]. Cummins and Gammon later modified the equation to account for the dispersion of thermodynamic properties in liquids [11], but it still relies solely on thermal variables. Based on this definition, the Landau–Placzek ratio is independent of scattering angle, implying that Rayleigh and Brillouin scattering should exhibit the same angle dependence and share similar scattering cross sections.

It is insightful to review the scattering theory of Rayleigh scattering, and here we consider the simplest model. In the case of a spherically symmetric molecule, a dipole moment $\vec{p}$ is induced in the same direction as the incident light's polarization. The scattered light amplitude is then a projection of this dipole moment in the direction of observation. The intensity of light I_s scattered from a single oscillating dipole is given as follows [14]:

$$I_s = \frac{\pi^2 c \vec{p}^2 \sin^2 \phi}{2\epsilon_0 \lambda^4 r^2}, \quad (2)$$

where ϕ is the angle of observation with respect to that dipole vector, c is the speed of light in vacuum, ϵ_0 is the vacuum dielectric constant, λ is the wavelength, and r is the distance between the observation point and the dipole.

From Eq. (2), we can infer a relationship between the angle and polarization dependencies. For maximum efficiency, the polarization direction of the incident light should be perpendicular to the scattering plane. When the incident light is circularly polarized, the scattering intensity is proportional to $1 + \cos^2 \theta$, which can be derived by splitting the signal into two orthogonal polarization states. In this case, maximum efficiency is achieved in backward scattering. These conclusions also apply to Brillouin scattering.

Experimental setup. To validate the theoretical predictions, we measured the Landau–Placzek ratios for four different liquids across various scattering angles. A schematic of the off-axis Brillouin spectroscopy setup is shown in Figs. 1(a)–1(b). A single-frequency external-cavity diode laser at 780 nm (TA-Pro, TOPTICA) was used as the light source. To calculate the scattering angle more accurately, the input beam size was set to 2.1 mm ($1/e^2$ intensity diameter), collimated by a fiber collimator. The pump light was linearly polarized and controlled by a half-wave plate, with the specific polarization angle adjusted according to the requirements of each experiment. Two mirrors were employed to adjust the incident light, ensuring it was centered on an achromatic lens and parallel to its optical axis. The lens was mounted on a three-axis stage with a precision of 0.5 μ m for tuning, allowing lateral displacement to vary the scattering angle. The angle was determined by the focal length of the lens and the off-center distance, with the tunable range constrained by the lens's NA. In this setup, the pump beam was incident on one side of the lens, and the scattered beam was collected on the opposite side, forming an off-axis geometry. The scattered light was selected using

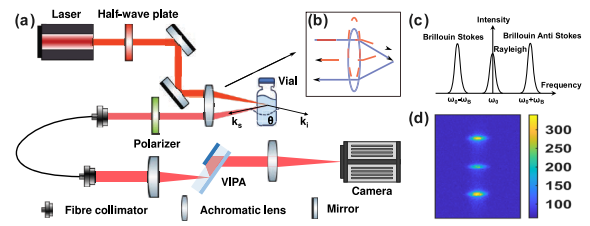


Fig. 1. (a) Schematic of the off-axis Brillouin spectroscopy. k_i is the incident light wave vector, and k_s is the scattered light wave vector, separated by the scattering angle θ . VIPA, virtual imaged phased array. (b) Diagram of the angle adjustment mechanism, where the red dashed line and blue solid line represent two scattering geometries. (c) Illustration of the Rayleigh–Brillouin spectrum in liquids, with most stray light eliminated. (d) Representative Rayleigh–Brillouin spectrum captured by the camera, showing the Rayleigh signal at the center.

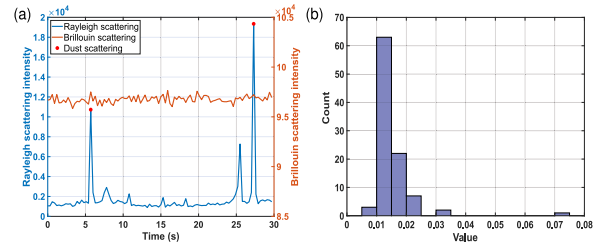


Fig. 2. (a) Rayleigh and Brillouin intensities measured over 30 s. Red dots represent dust scattering and are filtered during post-processing. (b) Histogram of Landau–Placzek ratios for water, based on 98 spectra (excluding two abnormal points).

a polarizer, maintaining the same polarization direction as the pump light. It was then coupled into a single-mode fiber (acting as a confocal pinhole) and directed to a VIPA spectrometer. Inside the spectrometer, a cylindrical lens was used to line-focus the collimated beam, which entered the VIPA at a slight angle. The spectrally dispersed light was captured by a scientific complementary metal–oxide–semiconductor (sCMOS) camera, forming a two-dimensional image (Fig. 1(d)).

Measurement of the Landau–Placzek ratio for water.

To test the system's performance, we first measured the Landau–Placzek ratio for water and compared the results with those from previous studies. The sample was filtered using a 0.22 μ m syringe filter to remove dust particles as much as possible and placed in a cylindrical quartz vial. We found that the empty vial produced negligible scattering, ensuring that most stray light was eliminated. Due to the low Rayleigh scattering intensity in water, we set the input light power to 184 mW and the camera exposure time to 300 ms to achieve a sufficient signal-to-noise ratio. We then selected a 0.40 NA lens with an 8 mm off-center displacement and set the polarization state perpendicular to the scattering plane (s-polarization) to maximize scattering efficiency. We chose to conduct our measurements at a temperature of 296.15 K, collecting 100 spectra continuously over 30 s, as shown in Fig. 2.

We selected regions of interest in the image corresponding to Rayleigh and Brillouin scattering and integrated the pixel values in those areas to obtain the intensities, with Stokes and anti-Stokes intensities combined (see Supplement 1 for data collection and processing). Each point in Fig. 2(a) represents the

Table 1. Landau–Placzek Ratio of Water

Authors	Temperature (K)	Experiment	Theory
Connor and Schlupf [15]	298.15	0.025	0.01
Maisano <i>et al.</i> [16]	295.48	0.02	
Cummins and Gammon [11]	293.15	<0.02	0.008
Zykova <i>et al.</i> [17]	293.8	0.03	
Our work	296.15	0.015	

total intensity, with Brillouin scattering being approximately two orders of magnitude stronger than Rayleigh scattering. Obvious fluctuations are observed in the Rayleigh signal (highlighted in red), which are absent in the Brillouin signal. These fluctuations are attributed to Rayleigh scattering from tiny dust particles and were filtered out using the three-sigma rule. To minimize artifacts, only the most significant outliers were excluded. Figure 2(b) shows the distribution of the Landau–Placzek ratios over the 98 spectra (excluding two abnormal points). Over 60% of the ratios fell within the range of 0.01–0.015, with a minimum value of 0.0092, reflecting minimal contributions from dust and stray light. Although one remaining outlier falls within the 0.07–0.08 range, its impact on the final result is negligible, as the Landau–Placzek ratio was averaged over all spectra. The final mean value of 0.015 demonstrates the stability of our system.

The results from previous studies are listed in Table 1. Theoretically, the Landau–Placzek ratio for water is approximately 0.01 at around 296 K, but experimental results from previous studies have reported much higher values due to limitations in their systems. Additionally, water is prone to picking up and retaining dust particles [11,15], further complicating accurate measurements.

In a recent study, a scanning Fabry–Perot interferometer has been used with a single spectrum acquisition time of 10 s [17]. Despite assuming the dust scattering contribution of the higher Landau–Placzek ratio and only calculating the result within a 15% interval from the lowest values over 50 spectra, they still obtained a substantially higher ratio of 0.03.

In contrast, our system offers both higher acquisition speed and measurement accuracy. By measuring Rayleigh and Brillouin scattering simultaneously, laser power fluctuations are eliminated, and dust scattering can be easily identified. The faster acquisition time also allows for multiple measurements, which substantially reduces uncertainty from abnormal fluctuations.

Angle dependence experiments. To better demonstrate that Rayleigh and Brillouin scattering exhibit the same angle dependence, we measured the Landau–Placzek ratios of another three different liquids—methanol, ethanol, and acetone—at varying scattering angles, following the same processing method as described above. Minimizing stray light requires sufficient spatial separation between the incident and scattered light. However, strong spherical aberration distorts the scattered light as the off-center distance increases, hindering fiber coupling into the VIPA spectrometer. To address both the stray light and spherical aberration issues, we selected a 6–8 mm off-center distance, which is suitable for a lens with a 25 mm diameter. Three achromatic lenses with NA values of 0.17, 0.40, and 0.83 were used, achieving a wide scattering angle range from 115.5° to 165°. The polarization of the pump light was aligned parallel to the scattering plane, ensuring that we measured different Brillouin

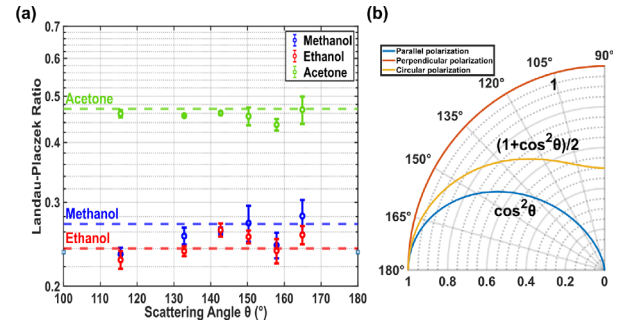


Fig. 3. (a) Landau–Placzek ratios for methanol, ethanol, and acetone at varying scattering angles. Dashed lines indicate theoretical values predicted by Cummins and Gammon [11]. (b) Derived angular distribution of Brillouin scattering intensities for three pump light polarization states, within the 90–180° scattering angle range (normalized intensities).

scattering coefficients for various scattering geometries. The results are shown in Fig. 3(a), where each data point represents the average of three discontinuous measurements, with each measurement consisting of 100 spectra over 20 s. For each data point, the standard deviations were calculated based on three measurements and were negligible for most cases. The Landau–Placzek ratios obtained at different scattering angles are consistent. The mean values across all scattering angles for methanol, ethanol, and acetone are 0.259 ± 0.017 , 0.246 ± 0.012 , and 0.461 ± 0.013 , respectively, which align well with the values predicted by Cummins and Gammon [11]. Although some deviations were observed between the results obtained in different scattering geometries, we believe these arise from the impact of the Gaussian envelope of the VIPA (see Supplement 1 for the Gaussian envelope) and have little effect on our conclusion. Furthermore, based on the previous analysis, we plotted the derived angular distribution of Brillouin scattering intensities in Fig. 3(b) for three pump light polarization states. Notably, the scattering intensity for circular polarization can be divided into two distinct parts.

Polarization dependence experiments. To validate the applicability of the dipole scattering model to Brillouin scattering in liquids, we examined the relationship between Brillouin intensity and polarization angle derived from this model. First, we confirmed that Brillouin scattering for methanol maintains its polarization state. Then, we rotated the half-wave plate to change the polarization direction of the input pump light and recorded the scattered light intensities through a polarizer aligned with each corresponding direction. The step size was set to 10°, and measurements were taken over a 180° range, with each angle measured ten times. Figure 4 shows that the Brillouin scattering intensity of methanol changes periodically, with a maximum observed at the s-polarization state, as previously analyzed.

Suppose the incident and scattered light wave vectors are $\vec{k}_i = (0, -1, 0)$ and $\vec{k}_s = (\sin\beta, \cos\beta, 0)$. The polarization of the incident light is $\vec{E}_i = (\sin\psi, 0, \cos\psi)$, where $\vec{E}_i$ lies in the x–z plane. Since $I_B \propto \sin^2\phi$ and $\phi = \cos^{-1}[\vec{E}_i \cdot \vec{k}_s]$, the scattering intensity can be expressed as follows:

$$I_B \propto 1 - \sin^2\beta \sin^2\psi, \quad (3)$$

where ψ represents the polarization direction of the pump light and β is the incident angle. This shows a clear relationship

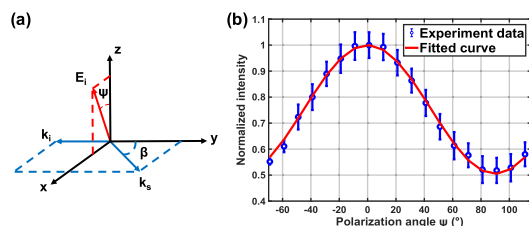


Fig. 4. (a) Schematic of the scattering geometry, where the scattering plane is in the x - y plane and the polarization of the incident light lies in the x - z plane. (b) Brillouin scattering intensity at different polarization angles. The intensity is normalized to the maximum scattering intensity, and the error bar is the standard deviation of ten measurements. The data is fitted based on Eq. (3).

between the Brillouin intensity and polarization angle. In backward scattering geometry, where $\beta = 0^\circ$, the scattering intensity is insensitive to the polarization state of the pump light. In right-angle scattering geometry, where $\beta = 90^\circ$, the scattering intensity vanishes when $\psi = 90^\circ$ (p-polarization).

In this experiment, we used a 0.83 NA lens with an 8 mm off-center displacement, and the incident angle β was calculated to be 47.8° (taking refraction in methanol into account). As shown in Fig. 4(b), the normalized intensities were fitted with Eq. (3), yielding a fitted incident angle of 44.7° . The slight deviation between these two angles arises from the rough approximation in the geometrical calculation, as we only considered the central light wave vector. Nevertheless, the results align well with our prediction, confirming that this simple dipole scattering model is suitable for Brillouin scattering in liquids.

In summary, we have demonstrated the angle-dependent intensity of Brillouin scattering in liquids, both theoretically and experimentally. We utilized the dispersion properties of VIPA in combination with an off-axis optical system that enables tunable scattering angle and polarization selection to measure the Landau–Placzek ratio, achieving faster acquisition times and higher precision. These advantages allowed us to conduct a series of measurements at various scattering angles. We measured the Landau–Placzek ratio of water as 0.015, improving accuracy over previous reports and approaching the theoretical value of 0.01. Through comprehensive analysis, we demonstrated that Brillouin and Rayleigh scattering exhibit the same angle dependence and share similar scattering cross sections. Additionally, polarization-dependent measurements validated the dipole scattering model for Brillouin scattering in liquids, providing a simple yet robust framework to understand its properties.

These findings not only enable the accurate calculation of the entire spectrum across arbitrary scattering geometries,

addressing a gap in previous work that primarily focused on the angular distribution of the q vector [8,18], but also highlight the importance of polarization in Brillouin scattering. Specifically, we noticed that Mattarelli *et al.* measured the scattered components from ballistic and diffusive photons in two polarization states [18], which led to varying scattering efficiencies and introduced deviations. Moreover, given the close relationship between Brillouin and Rayleigh scattering, the established research on Rayleigh scattering can provide valuable insights for Brillouin spectroscopy and microscopy.

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Data availability. Data underlying the results presented in this paper are not publicly available at this time but may be obtained from the authors upon reasonable request.

Supplemental document. See Supplement 1 for supporting content.

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